

Module 1: Work Order Management

Tab 7: Civil

1. Overview

The Civil tab is activated only after the **Excavation** phase is marked as complete. It manages all civil works, including daily forms, parallel lab testing, and the final "Before & After Verification" process. This tab is a critical step for restoring the site to its original condition after excavation and electrical work are finished.

2. Workflow

The Civil tab follows a structured workflow:

- 1 **Team Assignment:** The Project Manager (PM) or Civil Manager assigns one or more predefined Civil teams to the Work Order.
- 2 **Task Assignment Request:** The system checks the validity of the Task Number from the Excavation phase. If it has expired (older than one month), a new request must be generated.
- 3 **Daily Forms:** The system generates daily instances of Form - Contractor Team work and Form - Safety meeting.
- 4 **Lab Testing (Parallel):** The Civil Manager initiates and tracks lab tests for Sand, Concrete, and Asphalt through the external lab platform.
- 5 **Work & Photo Documentation:** The Civil team performs the work (Sand, Concrete, Asphalt) and captures photos at each stage, following the path defined in the Site Survey.
- 6 **Before & After Verification:** After all civil work is complete, the system guides the team to recapture photos from the exact same locations and angles as the initial "pre-work" photos taken during the Site Survey.

3. Team Assignment

- **Field:** Assigned Civil Team(s)
- **Data Type:** Multi-select dropdown
- **Source:** Predefined list of Civil teams from the HR Module.
- **Notes:** The PM or Civil Manager selects the team(s) responsible for the civil work.

4. Task Assignment Request

- The system first checks the issue_date of the task_assignment_requests record associated with the Excavation phase.
- If current_date - issue_date > 30 days, the system flags the Task Number as expired.

- A "Request New Task Number" button will be enabled, which will generate a new Form - Task Assignment Request using the same data as the original.
- **Field:** Task Number
- **Data Type:** Integer
- **Notes:** This field will display the valid Task Number. If a new one is requested, this field will be updated upon receipt.

5. Daily Forms

- The system will automatically generate daily instances of the following forms, which can be accessed from a list view within this tab:
 - Form - Contractor Team work
 - Form - Safety meeting

6. Lab Testing

This section runs in parallel with the main civil work. It allows the Civil Manager to manage the entire lab testing lifecycle.

6.1. Lab Management

- **Field:** Authorized Lab
- **Data Type:** Dropdown
- **Source:** A list of approved labs, managed in the System Configuration module.
- **Notes:** The Civil Manager selects the lab for the current project.

6.2. Test Request Management

- A table will list all test requests for the Work Order.
- **Columns:** Test Type, Request Date, Status, Report.
- A "New Test Request" button will open a form to create a new request, pre-populated with data from the Work Order.
- The system will provide a link to the external lab platform (<https://online.almustwa.com>) for submitting the request.

6.3. Test Stages

The following test stages will be managed:

- **Sand Test:**
 - Photo documentation of the sand being delivered and used.
- **Ready-mix Concrete Testing:**
 - Photo documentation of the concrete pour.
- **Asphalt Test:**

- Photo documentation of the asphalt being laid.

7. Before & After Verification

This is the final step of the Civil phase.

- The system will display a map showing the GPS coordinates of each photo taken during the **Site Survey** (specifically, the "Photos prior to site safety preparations" group).
- When the user selects a point on the map, the system will display the original "before" photo on one half of the screen.
- The other half of the screen will activate the device camera, allowing the user to take a new "after" photo.
- The system will provide visual guides (e.g., overlays) to help the user match the angle and framing of the original photo.
- Once all "before" photos have a corresponding "after" photo, this stage is marked as complete.

8. Completion

- Upon the successful completion of the **Before & After Verification** stage, the Civil tab is marked as complete.